

Reproduction of rasbt/LLMs-from-scratch: GPT-2 fine-tuning for spam classification

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Independent reproduction · Other · Natural language processing

Abstract

Ran the chapter 6 script `gpt_class_finetune.py` unmodified at the pinned commit: loads the pretrained GPT-2 124M weights, replaces the output head with a two-class classifier, and fine-tunes on the UCI SMS Spam Collection for 5 epochs with AdamW (lr 5e-5), reporting training and validation loss and accuracy. One NVIDIA RTX 5060 Ti.

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	GPT-2 spam classification fine...	2026-10-07	1	1	Signed by the K-Veritas server

2 Results

r1: GPT-2 spam classification fine-tuning

Run	Metric	Final value	Step
1	train_loss	0.083	600
1	val_loss	0.074	600
1	train_accuracy	100	5
1	val_accuracy	97.5	5
1	training_minutes	0.32	

Run	Inline claim	Value	Phase
1	final_train_accuracy	100	
1	final_val_accuracy	97.5	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	28.5 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	d2b8fc3c98a95d94c08f151967f8c35de473ca57708ecd37c37c0fe2c0ef9f5e	513 KB
r1.kvbundle.zip	c48ba9b0e18a012a4ec84ffd175e42d62e13f1ecf9979d7a4a2a5631760909c4	2.0 MB

5 Links and licence

Reproduction of: <https://github.com/rasbt/LLMs-from-scratch/tree/cdbd33e6d57a71e20ea31434841095fac50cecd8>

Record: CC-BY 4.0. Code: Apache-2.0, as declared by the authors. Tags: gpt-2, fine-tuning, text-classification, llm.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (GPT-2 spam classification fine-tuning), kv:2610.00012v1/r1. Data hash 0e679d2750be501506cdce3c313cf4c7517b8764e6495f6425cfc40170929ce8.

<https://kveritas.org/records/2610.00012v1/r1.pdf>